Maitreyi Muthya

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(Open to Relocate)

SUMMARY

Results-driven **Developer** with **4+ years** of experience designing and developing scalable, high-performance applications. Proficient in **backend (Java, Python, Node.js, C++)** and **frontend (React.js, JavaScript, TypeScript)** development. Skilled in **cloud computing (AWS, Docker, Kubernetes)** and **DevOps (CI/CD, Jenkins)** to optimize deployments. Proven track record in **API** performance optimization **(-35% response time),** monolith-to-microservices migration**,** and scaling systems **(10K+ users)**. Strong expertise in **FinTech, payment integrations,** and data-driven solutions. **Actively seeking opportunities** to apply analytics expertise in driving innovation and growth strategies

AWARDS & RECOGNITION

- Living Proof Award at Fiserv for outstanding client service achievement, process improvement initiatives, and exceptional monthly productivity
- Shining Star Award (Q1 & Q2 2021) at Fiserv for backend optimization, API development, and performance enhancements

WORK EXPERIENCE

RPG RESEARCH — **Remote, US**

Developer Jul 2024 - Present

Designed and optimized cross-platform game software, improving performance, scalability, and accessibility with BCI integration

- Directed technical design strategies and led cross-disciplinary teams in developing cross-platform game software with C++, JavaScript, and **Godot**, improving scalability and maintainability
- Optimized game engine performance by debugging and tuning game systems, reducing loading times by 25%, and enhancing user experience
- Implemented **Python**-based AI modules, enhancing NPC behavior and improving player engagement by 20%, boosting realism
- Designed **SQL** database schemas for character data, enabling real-time synchronization and seamless data persistence
- Deployed applications using cloud services (**AWS, GCP, Azure**) with advanced DevOps practices.
- Integrated Brain-Computer Interface (**BCI**) technology for accessibility, increasing market reach by 15%
- Developed a character data management system in **C++** and **JavaScript**, enabling real-time synchronization, seamless data persistence, and improved game state consistency, reducing data inconsistencies by 30%
- Built a dynamic UI-driven character customization system using **JavaScript, HTML5,** and **CSS**, integrating modular design principles with real-time attribute modifications and seamless game state updates, enhancing accessibility and gameplay responsiveness

FISERV — **Pune, IND**

Software Engineer — **Oct 2019 - June 2022**

Project 1: Fiserv University - Internal Education Platform

Refactored Fiserv's internal learning platform by converting the monolithic backend to a microservices architecture, implementing caching, and optimizing API calls to resolve slow response times, database bottlenecks, and scalability issues. Improved system performance enabled the platform to scale beyond 10K+ daily users, supporting internal training and external client certifications.

- Modernized internal and client-facing platforms at Fiserv by building scalable backend services with Node.js and Python, and revamping frontends using **React.js** and **Redux** Toolkit. Improved performance, reliability, and user experience across education and digital payment systems, impacting **10K+** users.
- Engineered a scalable learning platform for onboarding, training, and certification of Fiserv employees and external clients, enhancing workforce upskilling and product education using **Node.js, Python** (Flask), and **MySQL**
- Optimized platform performance, enabling it to scale from 5K to 10 K+ daily active users by redesigning the backend from a legacy **PHP** & **SQL** stack to a microservices architecture using Node.js, Kubernetes, **Docker**, and MySQL

- Refactored inefficient legacy code, eliminating nested loops, redundant API calls, and poor sorting algorithms by leveraging hash maps and optimized sorting techniques, reducing computation overhead by 20% and improving data processing speeds
- Redesigned the database schema, replacing denormalized structures with a normalized **MySQL** schema and adding indexes, improving query performance, reducing data redundancy by 20%, and decreasing course loading times.
- Developed and optimized **RESTful** APIs with **Python** (Flask) and **Node.js** (Express.js), implementing efficient routing and Redis-based caching, reducing API response times by 35% and offloading excessive database queries
- Automated **CI/CD** pipelines using Jenkins, GitHub Actions, and Kubernetes, reducing deployment time by **50%** and ensuring faster, more reliable feature rollouts with zero downtime updates.
- Expanded platform usage beyond internal employees—after launching the optimized system, leadership extended it to external clients for certification programs, validating the platform's success as a scalable and industry-leading learning solution

Project 2 :
Client: Blue Shield of California, US
Integrated Fiserv's Engagement Advantage into BSCA's outdated payment portal, enabling modern payment methods and enhancing performance, usability, and member engagement.

- Migrated legacy frontend from **AngularJS**/jQuery to **React.js (Next.js)**, significantly improving application performance, scalability, and SEO, and laying the foundation for future enhancements
- Integrated Fiserv's Engagement Advantage **APIs**, enabling modern and secure payment methods, including ACH, credit/debit cards, and digital wallets (Apple Pay, Google Pay, PayPal), enhancing user payment flexibility
- Implemented efficient state management using Redux Toolkit and React Query, reducing redundant API calls and improving frontend responsiveness and data handling
- Established automated **CI/CD** pipelines with Jenkins and GitHub Actions, streamlining deployment processes and reducing release downtime by 50%

PROJECTS

WeatherJag - Live Weather Tracking Application

- Developed a real-time weather tracking application using Python and PHP, integrating the OpenWeatherMap API for accurate forecasts
- Optimized MongoDB database structure, reducing data retrieval latency by 30%, improving app performance

Hypernymy Detection - Machine Learning Benchmarking

- Implemented an SVM-based hypernym detection model using Python, TensorFlow, and NLTK, achieving 83% accuracy, 75% precision, and 68% F1-score
- Outperformed Deep Neural Networks and XGBoost, demonstrating superior efficiency in hypernym detection tasks

Stock-Prediction-through-news-sentiment-analysis

- Built a Python-based stock prediction model using Yahoo Finance data and New York Times news sentiment, spanning 10 years
- Trained a Random Forest Regressor, achieving 84% accuracy, and analyzed prediction performance with and without smoothing
- Identified that ensemble averaging caused underestimation of extreme price swings, leading to further exploration of feature engineering and alternative models (XGBoost, LSTMs)
- Visualized stock trends, comparing aligned vs. raw predictions, and optimized temporal alignment to improve market sentiment correlation

SKILLS

- **Programming Languages:** C++, Java, Python, JavaScript, TypeScript, SQL, PHP
- **Databases**: MySQL, PostgreSQL, MongoDB
- **Backend & API Development:** Node.js, Django, REST APIs, Postman
- **Frameworks & Libraries:** React, Express, Flask, Bootstrap, jQuery, pandas, NumPy
- **Cloud & DevOps:** AWS (Lambda, S3, EC2), CI/CD, Kafka

- **Containerization & Deployment:** Docker, Kubernetes, Jenkins
- **Other Tools**: Git, JIRA, Jenkins, PowerBI, Tableau

EDUCATION

PURDUE UNIVERSITY **Indianapolis, USA**
MS in Computer & Information Science (GPA: 3.5/4) **May 2024**

PUNE UNIVERSITY **Pune, IND**
B.E in Information Technology **May 2019**